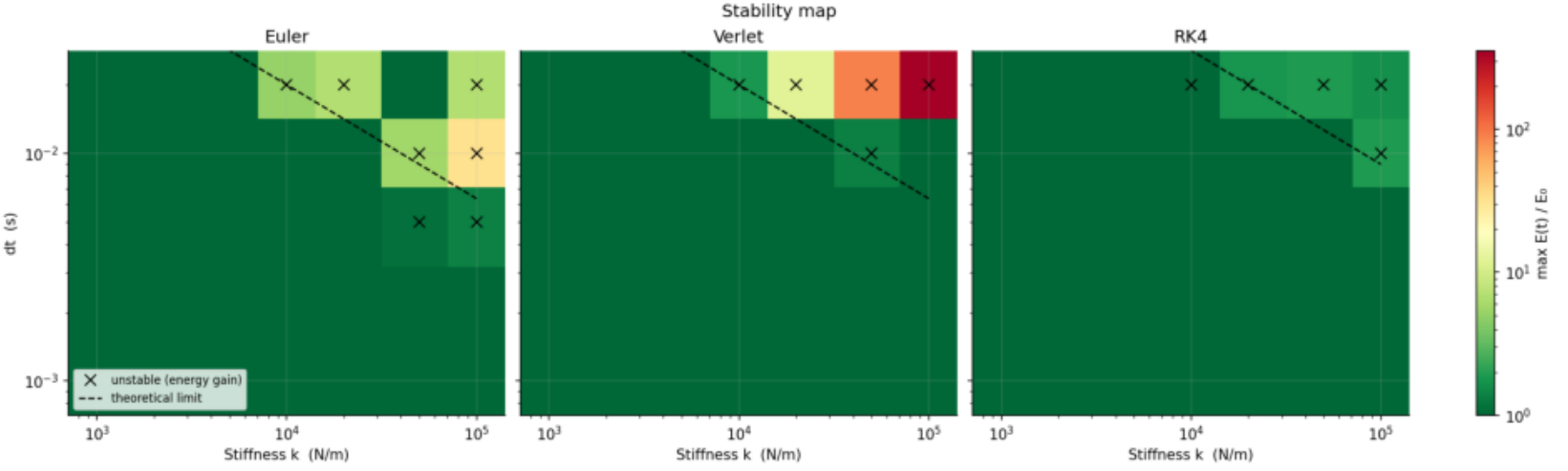
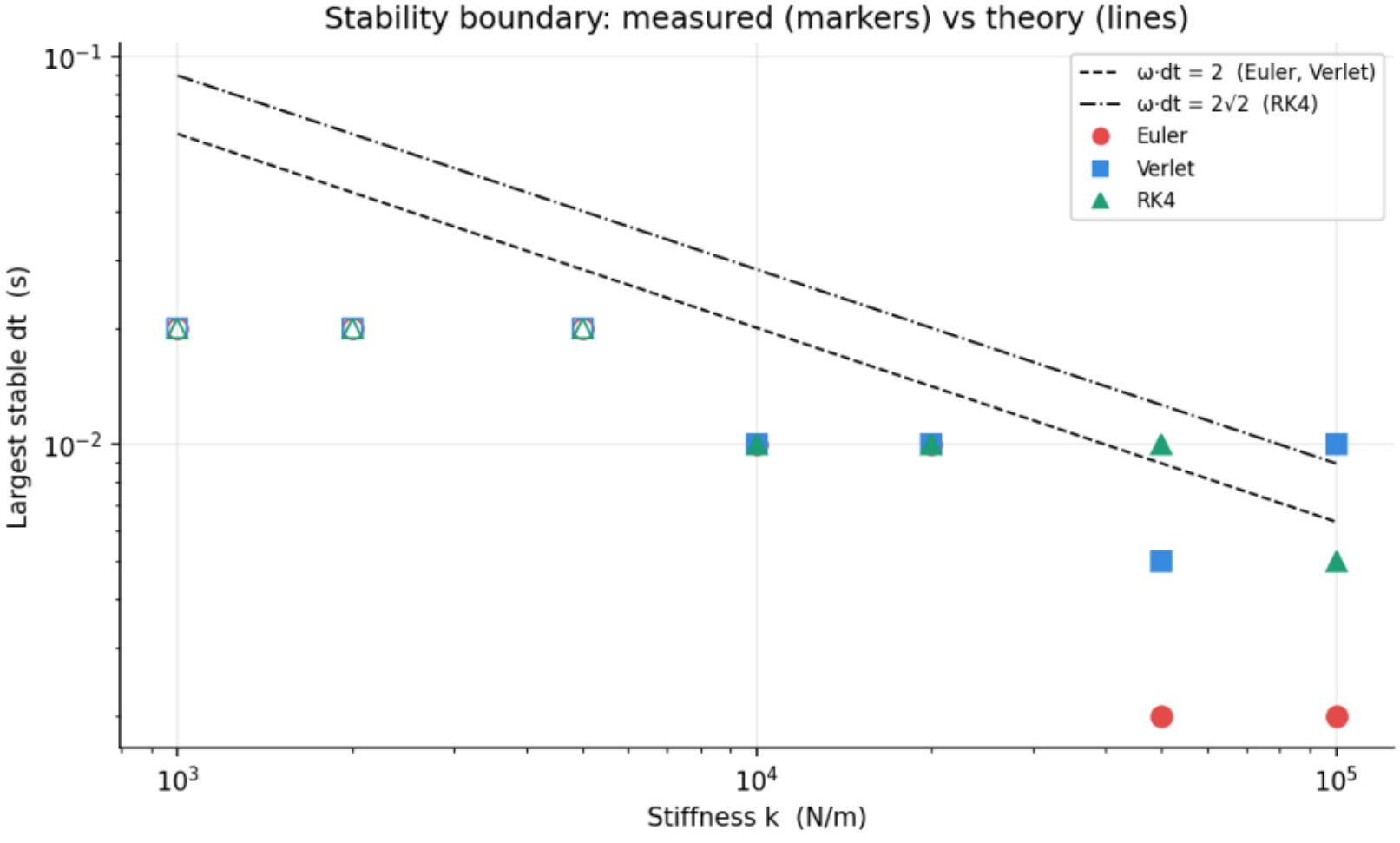
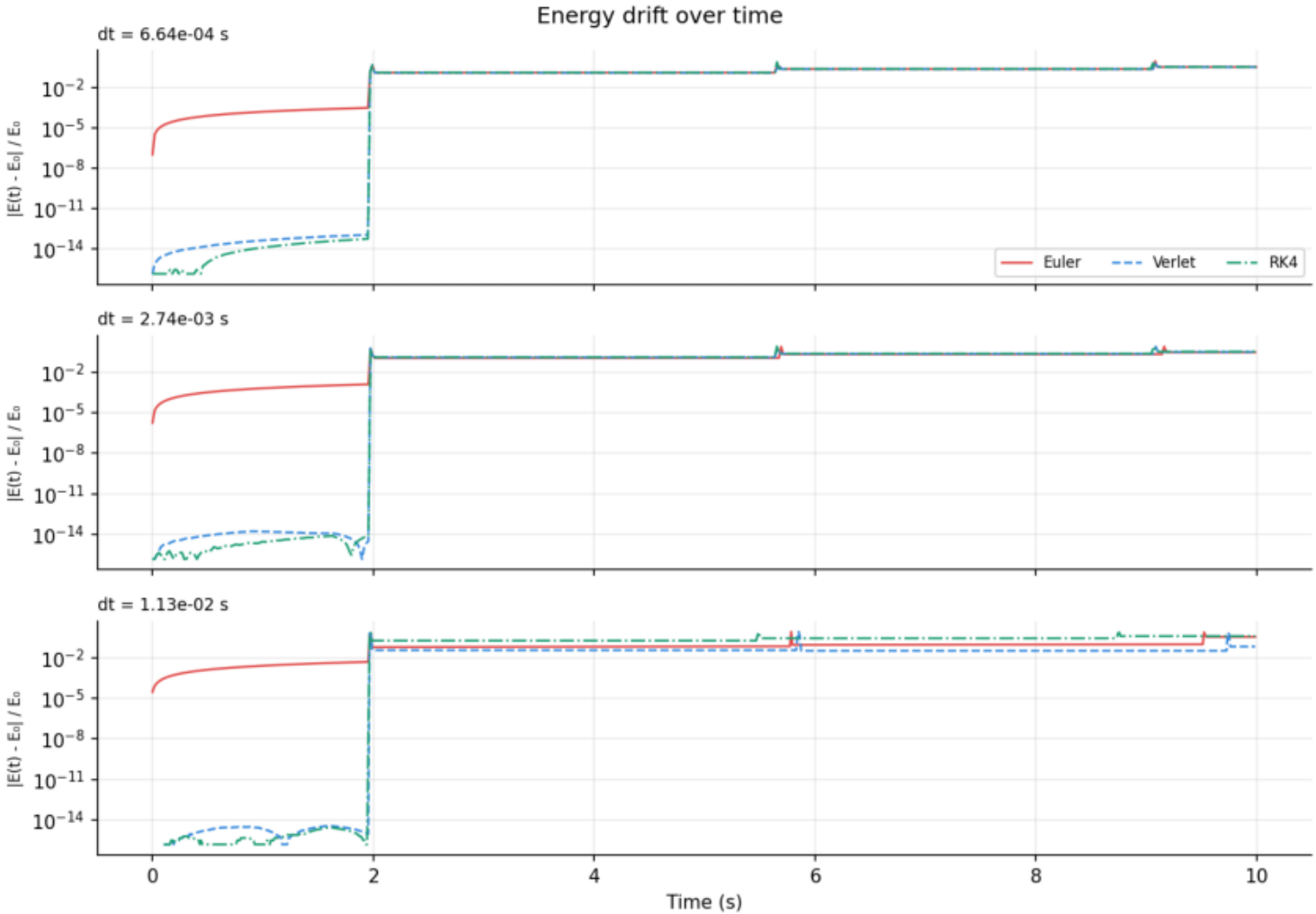

Peak height error vs dt
Euler
Verlet
RK4
Euler: slope 1
Verlet: slope 2
RK4: slope 4
max relative peak height error (vs reference run)
dt (s)
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-3}
10^{-2}


Stability map
Euler
Verlet
RK4
dt (s)
10^{-2}
10^{-3}
10^{3}
10^{4}
10^{5}
Stiffness k (N/m)
unstable (energy gain)
theoretical limit
max E(t) / E₀
10^{2}
10^{1}
10^{0}


Stability boundary: measured (markers) vs theory (lines)
ω·dt = 2 (Euler, Verlet)
ω·dt = 2√2 (RK4)
Euler
Verlet
RK4
Largest stable dt (s)
Stiffness k (N/m)
10^-1
10^-2
10^3
10^4
10^5


Energy drift over time
dt = 6.64e-04 s
dt = 2.74e-03 s
dt = 1.13e-02 s
|E(t) - E₀| / E₀
10⁻²
10⁻⁵
10⁻⁸
10⁻¹¹
10⁻¹⁴
Euler
Verlet
RK4
0
2
4
6
8
10
Time (s)